

LOCATE: A LoRa-based mObile emergenCy mAnagement system

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Abstract—During the occurrence of an emergency, being it the consequence of a natural disaster or of human activities, the pervasiveness of user-owned mobile devices (e.g. smartphones, tablets) turns into a precious help to convey data and services to all the people involved. As a result, several emergency-related mobile applications have been proposed on the market; however, they are limited by the networking capabilities of the devices, since they are often based on short-range Device-to-Device (D2D) communication technologies (e.g. the Wi-Fi Direct), or on the cellular infrastructure, which might be not available on the emergency scenario. In this paper, we attempt to overcome both such issues, by proposing a novel Emergency Communication System (ECS) which operates over infrastructure-less phone-based networks, and guarantees long-range D2D communication thanks to the LoRa technology. The system, named **LOCATE**, includes a mobile application, through which users can convey minimal yet vital emergency-related data, and a dissemination protocol, spreading the emergency requests over multi-hop LoRa links. The performance of **LOCATE** have been evaluated through: (i) an experimental study, assessing the capability of LoRa technology to convey short, emergency messages over long distances, and a (ii) simulation study, demonstrating the effectiveness of the dissemination protocol on large-scale scenarios when compared to state-of-the-art flooding schemes.

I. INTRODUCTION

Emergency Communication Systems (ECS) [1] play the crucial role of enabling information exchange among individuals and rescue teams in critical situations where traditional infrastructures (e.g. cellular networks) are not physically available, like in rural areas, or temporally unavailable, like in post-disaster scenarios. Being context-awareness and self-adaptiveness two fundamental requirements of ECS, several studies have investigated the utilization of multi-hop Mobile Ad Hoc Networks (MANETs), but with few real deployments [2]. More recently, the worldwide diffusion of end-user mobile devices, like smartphones and tablets, has paved the way toward next-generation ECS able to leverage on the pervasiveness of such devices, as well as on the presence of embedded sensors (like the GPS) providing context-information about the emergency and the user [3]. Unfortunately, most of smartphones do not support the ad hoc mode for security reasons; however, the recent models are equipped with Device to Device (D2D) communication technologies, enabling data exchange with other peers in the 1-hop neighborhood. Several emergency-related mobile applications based on Wi-Fi Direct (e.g. FireChat [4], Help Me [5], Emergency Direct [6]) as well as research studies on D2D communications over cellular bands have been proposed [7]. At the same time, it is worth

remarking that ECS based on short-range communications are only feasible on small-scale environments, while D2D solutions on cellular bands still require the support of the base-stations for device discovery and resource allocation [7].

In this paper, we make a step further on phone-based ECS, by taking advantage of the recent diffusion of Low Power Wide Area Networks (LPWANs) technologies [8], typically employed for the Internet of Things (IoT). In particular, we focus on LoRa [9], which is one of the most popular LPWAN solution currently available on the market. The LoRa scheme is a proprietary PHY layer solution; on top of it, the LoRaWAN standard defines the MAC layer and the network architecture, which allows connecting the LoRa devices to the Internet via intermediate gateways. The literature on LoRa systems is quite recent, and focuses on experimental evaluation of LoRa links (e.g. [10], [11]) and of LoRaWAN networks (e.g. [12], [13]). The multi-hop communication between LoRa transceivers (i.e. without intermediate gateways) is barely addressed; however, recent studies like [14] have demonstrated its feasibility thanks to the time- and frequency- domain energy spreading effect that greatly alleviates the synchronized collision problem.

In this paper, we present **LOCATE**, a phone-based ECS which allows multi-hop dissemination of emergency messages over LoRa links. The proposed system is composed by an Android application connected to a LoRa System on Chip (SoC) via a USB cable, as depicted in Figure 1. More specifically, we provide three main contributions in this paper:

- We propose an experimental evaluation of **LOCATE** devices, aimed to characterize the suitability of LoRa with respect to the ECS requirement, and to derive analytical models of the LoRa links. The experimental results demonstrate the possibility to disseminate emergency messages with minimal -but still adequate- payload, while guaranteeing a communication range up to 1 km, all by using COTS, low-cost equipment.
- Based on such results, we describe the **LOCATE** mobile application, through which the user can broadcast geo-tagged help requests on the LoRa link. The message is then re-broadcasted through a novel multi-hop dissemination scheme, until a node able to resolve the emergency is detected.
- We evaluate the proposed dissemination scheme on simulated, large-scale OMNeT++ scenarios, by tuning the LoRa model according to the experimental results. The

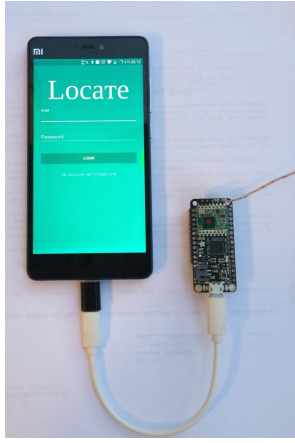


Fig. 1. The LOCATE mobile application and the LoRa module.

simulation study demonstrates that the proposed scheme is able to determine the best trade-off between emergency resolution time and network overhead when compared to other multi-hop forwarding schemes.

The rest of the paper is structured as follows. In Section II, we review the related literature on LoRa research. Section III provides experimental results for a LOCATE single-hop testbed. Section IV describes the LOCATE architecture and components, including the mobile app and the multi-hop message dissemination protocol. This latter is evaluated in Section V. Conclusions follow in Section VI.

II. RELATED WORKS

Although no direct applications of the LoRa technology to ECS have been proposed so far in the literature, we register an increasing amount of research studies on the LoRa IoT systems, thanks to the hardware cost reduction, and to the increasing availability of software platforms for the network deployment [15]. More in details, we can classify the existing LoRa-related studies in three main domains: (i) experimental test-beds, characterizing the performance of LoRa links and LoRaWAN networks on different scenarios; (ii) novel use-cases for LPWAN technologies, e.g. related to agriculture [16] or to smart-city environments [17], and (iii) enhancements to the MAC/PHY layers or to the network architecture [14] [18]. The first category of studies propose experimental measurements aimed to evaluate the LoRa technology with respect to performance, energy consumption or range of the network: for instance, the authors of [10] performed an extensive evaluation of LoRa coverage, and observed a maximum communication range of more than 15 Km on both ground and water scenarios. The scalability of LoRaWAN in dense scenarios has been investigated by means of theoretical framework and simulation studies respectively in [12] and [13]; in particular, this latter demonstrates that the goal of connecting hundreds of devices to the same gateway -while guaranteeing a high packet success rate- is feasible. A comprehensive analysis of how the packet loss of a LoRa link is affected by operative

conditions like the transmit power, the payload length and the antenna orientation has been provided in [19], while an empirical path loss model derived from signal measurements in the 433 MHz and 868 MHz ISM bands has been discussed in [11]. Among the several enhancements proposed in the literature, we cite the studies in [14] and [18] which address the problem of multi-hop LoRa networks ensuring D2D, any-to-any communications among LoRa transceivers. In particular, [18] proposes a novel MAC protocol allowing the selection of the Spreading Factor (SF) at each link; this solution is then coupled with the RPL routing protocol in order to determine the routing paths minimizing the end-to-end time-on-air metric. In [14], the authors describe a multi-hop flooding protocol that copes with the synchronized packet transmissions by employing a TDMA-based technique. Our solution starts from the findings in [14], although designing a different solution, which combines timing offsets and Delay Tolerant Networking (DTN)-based mechanisms in order to maximize the effectiveness of the dissemination process.

III. EXPERIMENTAL RESULTS

In this Section, we provide a preliminary experimental evaluation of the D2D LoRa communication through the smartphone-based equipment depicted in Figure 1. Despite the number of related works focusing on the experimental characterization of LoRa and LoRaWAN, our evaluation includes novel results aimed to: (i) derive proper statistical models and threshold values, which are used to tune the simulation model described in Section V; (ii) understand drawbacks and gains of LoRa technology for emergency-related applications, running on the COTS mobile devices of Figure 1 and executed on generic, uncontrolled environments. To this aim, we consider an Adafruit Feather M0 RFM96 LoRa Radio (433MHz), connected to an Android smartphone via the USB cable. We deployed a novel Android Application for testing purposes, which allows to: (i) tune the parameters of the LoRa transceiver, i.e. the Spreading Factor (SF), Bandwidth (BW), Transmit Power (TP), Coding Rate (CR), Packet Size (PS); (ii) inject network traffic according to a Constant Bit Rate (CBR) application model; (iii) compute network statistics for each experiment, like the mean delay, throughput, Packet Delivery Ratio (PDR); (iv) display the results on a map, including the GPS data. We ran multiple experiments for single-hop D2D communication, considering two mobile users walking on the urban scenario (the downtown of Bologna, Italy). Every λ seconds, the source device transmits a message, and waits to receive the corresponding application ACK.

Figure 2(a) depicts the PDR over distance, when considering the following configuration: $SF=128$ chips/symbol, $BW=125$ KHz, $CR=4/5$, $\lambda=3$ seconds, and varying the TP, until the maximum admissible value according to the Italian regulations (10dbm). Using such configuration, the maximum achievable range is around 600 meters, which is a quite lower value compared to the results in [10]. However, we should consider that: (i) tests have been performed by using low-cost antennas as the real-world deployment in mobile emergency

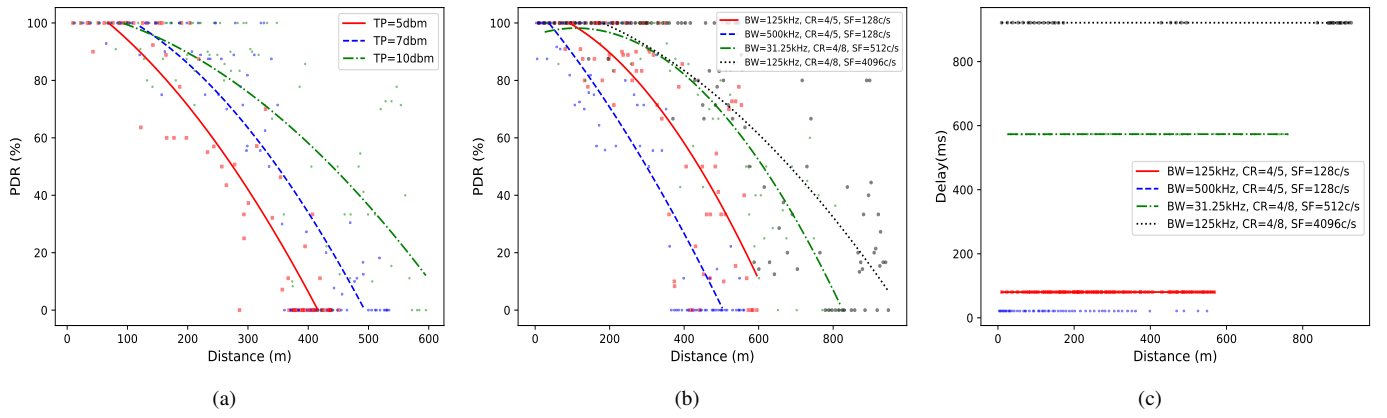


Fig. 2. Experimental characterization of the LoRa communication through the devices of Figure 1: the PDR for different TP values, and for different BW , CR , SP configurations is shown in Figure 2(a) and 2(b), respectively. The delay for different configurations is shown in Figure 2(c).

scenarios; (ii) in any case, LoRa range is considerably larger than other D2D communication technologies used for ECS applications (e.g. the Wi-Fi Direct tested in [6]). In Figure 2(b) and Figure 2(c), we depict respectively the PDR and delay for different configurations of the SF , BW and CR parameters; it is easy to notice that, increasing the SF (i.e. 4096), the maximum range achieves the 1 Km, although the delay metric also increases. Similarly, in Figure 3 we depict the PDR and delay, for the same configuration of Figure 2(a) and different values of PS : as expected, the delay increases considerably with PS , while the PDR is almost not affected. From Figures 2 and 3, we can conclude that: (i) due to the high delay, emergency management with strict real-time requirements cannot be supported; (ii) similarly, the limited PS size does not allow implementing most of the emergency services proposed so far (e.g. chat like [4] [5]), however it is enough to support the exchange of minimal, yet vital information (e.g. the GPS coordinates); (iii) the range is up to ten times those of the Wi-Fi Direct, and hence message dissemination over a large-scale scenario could require only a limited number of retransmissions.¹

IV. LOCATE: ARCHITECTURE AND COMPONENTS

The LOCATE system provides multi-hop dissemination of emergency messages over generic, large-scale scenarios composed of mobile users, without assuming the presence of a fixed infrastructure, which can be missing (e.g. in rural environments), or temporarily not available (e.g. in a post-disaster environment). The proposed network architecture can be considered an instance of MANET, however introducing the following peculiarities: (i) it is composed by smartphone devices transmitting over LoRa links, by means of the equipment shown in Figure 1; (ii) it supports opportunistic message dissemination without any additional overhead for the topology control and network setup; (iii) it is constrained by the technology, in terms of traffic type (only broadcast

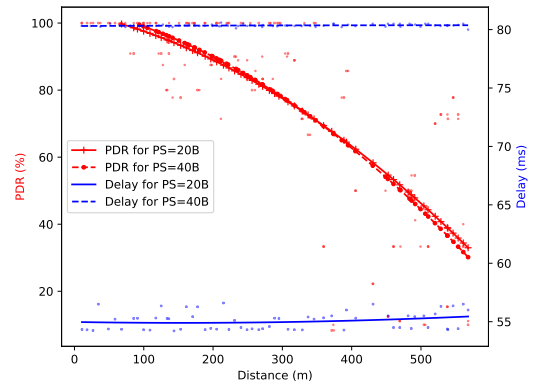


Fig. 3. The PDR and delay over distance for different values of LoRa packet size (PS).

communication is supported) and load (only short messages with minimal information can be transmitted, as discussed in Section III). At the same time, the LOCATE system does not require LoRaWAN gateways, and natively operates via multi-hop D2D communications between LoRa devices. The LoRaWAN MAC scheme is not employed, and the issue of synchronized collisions is handled through proper timing mechanisms at the application layer, although such risk is already mitigated by the PHY layer, as demonstrated in [14].

Figure 4 clarifies the main operations of the proposed ECS. We assume the presence of three kinds of users/nodes in the scenario: (i) emergency sources (ES), i.e. users needing some kind of help (e.g. requesting medical assistance), who will start an emergency procedure via the Mobile App; (ii) emergency solvers (EV), i.e. users having the expertise (e.g. medical personnel) or the possibility (e.g. having a feasible path toward the ES) to provide their help once notified; (iii) emergency relays (ER), i.e. users who are not able to act directly on the emergency, but can facilitate the search for a solver in their neighbourhood. We highlight that the previous roles are purely logical, i.e. the same user can serve as ES , EV , or ER at different instants, or for the different emergency

¹For instance, only three hops might be needed to cover the diameter of the downtown of Bologna, assuming a transmission range of 800 meters.

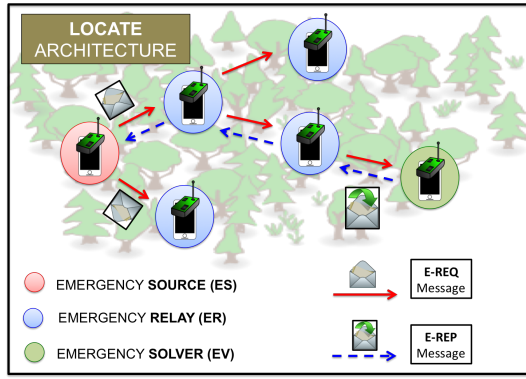


Fig. 4. The LOCATE network architecture.

requests.

Any node can become an *ES* by broadcasting an Emergency Request (E-REQ) message, containing the following fields:

$$\langle request_{id}, position, type, user_{id}, device_{id} \rangle \quad (1)$$

Here, $request_{id}$ is a 2 bytes sequence number; $position$ is a 8 bytes field containing the (latitude, longitude) coordinates of the *ES*, acquired through the GPS embedded within the smartphone; $type$ is a 1 byte field, characterizing the nature of the emergency (e.g. rescue help, medical help, ...) and the type of the message (E-REQ or E-REP); $user_{id}$ (2 bytes) is the unique user identifier, generated when downloading the app the first time; $device_{id}$ (6 bytes) is the MAC address of the transmitting LoRa device. Any user receiving the E-REQ message can: (i) accept to handle the request via the mobile app, hence becoming an *EV*; or (ii) serve as *ER*. The acceptance can be performed manually or automatically, depending on whether the corresponding preference of the mobile app has been turned on or off. Vice versa, the election to *ER* is always performed automatically; in this case, the E-REQ message is broadcasted according to the dissemination protocol described in Section IV-B, by replacing the $device_{id}$ field with the local MAC address. When becoming an *EV*, the node will broadcast an Emergency Reply (E-REP) message having the same structure of the E-REQ, except for the $type$ field. The E-REP message is automatically re-broadcasted by each *ER*, until reaching the *ES*. We do not further elaborate on the way the emergency is managed, since this is out of scope of this paper. In the following, we detail the application (Section IV-A) and the dissemination protocol (Section IV-B).

A. Mobile Application

The LOCATE Android Mobile Application supports two main functionalities: (i) it provides the GUI allowing to start a new emergency procedure, or to track the ongoing procedures started by other peers; (ii) it implements the multi-hop dissemination protocol described in the next Section. The communication with the LoRa node is implemented through the USB Host library offered by the Android OS. We remark that all the complexity of the dissemination process is implemented on the smartphone side, hence the LoRa radio simply



Fig. 5. Screenshots of the LOCATE Mobile Application. The register page is shown in Figure 5(a); the main operation are listed in Figure 5(b); Figure 5(c) shows how to change the settings for the LoRa module; finally, an emergency request is depicted in Figure 5(d).

transmits the packets received from the wireless medium to the Serial port, or the opposite (i.e. it transmits the packet received from the Serial port to the wireless medium). In Figure 5, we report some of the main Activities of the mobile app. The first time the user is opening the application, s/he needs to register on the service, receiving a unique $user_{id}$ (Figure 5(a)). Figure 5(b) contains the list of the main operations supported by the app, i.e.: (i) to notify an emergency, indicating the type of help requested, and automatically retrieving the GPS coordinates; (ii) to monitor the list of ongoing emergencies (not solved yet) on an off-line (pre-downloaded) map; (iii) to change the settings of the LoRa module (Figure 5(c)). In case an E-REQ message is received, and the automatic reply has been disabled, a notification is displayed (Figure 5(d)) showing the type of emergency and GPS coordinates of the *ES*. Then, the system waits to receive a user feedback (accept or reply) until a maximum interval. If no feedback is received, the device becomes an *ER*, and re-broadcasts the message. The networking component is implemented on a separate thread, allowing the Mobile App to be executed in background.

B. Dissemination Protocol

For easy of disposition, we describe the case where a single emergency is managed within the phone-network, although the LOCATE dissemination scheme is not affected by the presence of multiple *ES*.

Let $L = \{l_1, l_2, \dots, l_n\}$ be the set of the n user-owned smartphones equipped with the LoRa radio interface and available within the scenario. Figure 6 depicts the state chart containing all the possible states and transitions of the node handling the i -th emergency. The protocol works in two main phases: (i) a dual, biased contention phase in which nodes become aware of the emergency for the first time, and forward the corresponding E-REP or E-REQ messages, becoming an *EV* or an *ER* (based on whether they are able to solve the emergency or not); and (ii) a DTN-inspired phase in which each *EV* implements a store-and-forward mechanism, until receiving an E-REP message from one of its neighbor. The goal of the contention phase is twofold, i.e. to let the possible solvers handle the emergency as soon as possible or, if there

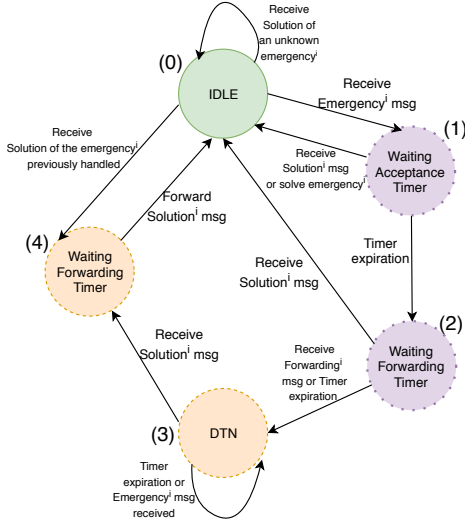


Fig. 6. The LOCATE dissemination protocol for the i^{th} -message. State 0 represents the initial state for a node. States 1 and 2 are the states corresponding to the emergency accepting and forwarding. States 3 and 4 are the DTN-based states through which the nodes keep the emergency alive over the network and possibly spread the solution.

are no solvers around, to make the E-REQ cover the longest distance without flooding the network. Let CW_{acc} be the acceptance contention window, such that:

$$CW_{acc} = CW_{max} \cdot \left(1 - \frac{1}{e^\delta}\right) \quad (2)$$

$$\delta = \gamma \cdot \frac{d(s, j)}{1 + \frac{d(s, j)}{r}} \quad (3)$$

where CW_{max} represents the asymptotic value for the delay, γ is a steepness factor, $d(s, j)$ is the distance between the emergency source l_s and the receiver node l_j , and r is the approximated maximum communication radius (the parameter is tuned according to the experiments reported in Section III). When receiving the E-REQ message for the first time, each node starts the *acceptance timer*, which is set to a random value uniformly distributed between 0 and CW_{acc} . It is easy to notice that nodes closer to the E-REQ source will have more chance to win the contention, since they might be able to reach the physical location of the *ES* quicker than the other nodes. If the nodes receiving the E-REQ did not send or overhear any E-REP message after a CW_{max} time interval, they start a second contention phase, aimed to detect the best candidate to forward the E-REQ message.

Let CW_{ft} be the forwarding contention window, such that:

$$CW_{ft} = CW_{max} \cdot \frac{1}{e^\delta} \quad (4)$$

The value of the *forwarding timer* is random and uniformly distributed between 0 and CW_{ft} . The formulation of the *acceptance timer* (Equation 2) and of the *forwarder timer* (Equation 4) is quite similar: however, the goals are opposite, since the *forwarding timer* gives higher priorities to nodes farther from the E-REQ transmitter, so that the emergency

is spread out as fast as possible. In the second phase of the protocol, after the *forwarding timer*, nodes enter in a DTN-like state till receiving an E-REP message. A node inside the DTN state regularly broadcasts the E-REQ message by waiting for a random value uniformly distributed between CW_{min} and CW_{max} , with $CW_{min} < CW_{max}$. Since several neighbor nodes can enter the DTN state and keep disseminating the same information, we adopt some precautions in order to limit the number of messages sent:

- A node in DTN state forwards a message with a probability equal to P_{DTN} , computed as follows:

$$P_{DTN} = (p_{start})^{\frac{1}{1+N}} \quad (5)$$

where p_{start} is the initial probability; its impact is analyzed in Section V.

- If a node waiting for its DTN delay overhears an E-REQ message for the same emergency from a neighbor, it reschedules the timer.
- If a node waiting for its DTN delay overhears more than one E-REQ message for the same emergency from N distinct neighbors, it freezes the retransmission timer, and restarts it after moving of at least DTN_{dist} meters.

As soon as a node in DTN state overhears an E-REP message, it starts a *forwarding emergency timer*, i.e., the contention phase previously described, in order to inform the *ES* as fast as possible. The procedure is started even in case the E-REP message refers to an emergency already marked as solved, in order to increase the chance to reach the *ES*.

V. PERFORMANCE EVALUATION

In this Section, we evaluate the performance of the LOCATE dissemination protocol over realistic, simulated emergency scenarios modeled through the *OMNeT++* tool, and compare it against other state-of-art flooding schemes. We rely on the FLORA module [20], which provides a fine-grained modeling of the LoRa PHY layer, and we properly tune the simulation environment with the experimental results presented in Section III (e.g. range, RSSI and PDR model).

A. Scenario

We consider a square scenario of $5km \times 5km$, with $n + 1$ LoRa nodes deployed over it. There is exactly one ES, which is assumed stationary and placed at the center of the scenario, while the other n nodes are randomly distributed over the area, and move according to a Random Direction mobility model, with random speed in range $[0.5:3]$ m/s (typical range of pedestrian mobility). Let τ represent the percentage of EV over the n nodes available. After a preliminary analysis, we tune the LOCATE parameters to the following values: CW_{max} (max delay) = 20s, γ (stepness) = 0.005, r (communication radius) = 500m. Furthermore, we used the following system parameters: CW_{min} (min delay) = 5s and DTN_{dist} (DTN radius) = 50m. We averaged each result over 1000 runs, in order to mitigate the impact of randomness. Each run ends when: (i) all the nodes who had received an E-REQ message marked the emergency as solved (i.e. they received

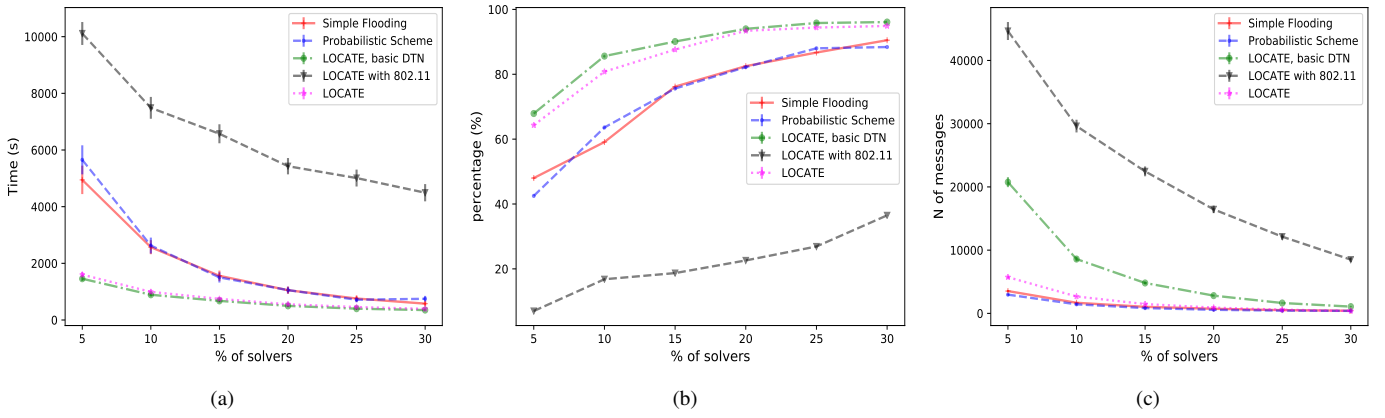


Fig. 7. The ERT , ERR and EO metrics for different values of τ and $n=40$ are shown in Figures 8(a), 8(b) and 8(c) respectively.

the corresponding E-REP) or (ii) the simulated time exceeds a maximum value, set to 1 day in the following. We neglect the emergency *type* in this analysis: we assume that the set of EV is pre-decided, and each time an EV receives any E-REQ message, it automatically replies with an E-REP message.

B. Other protocols

We compare our solution against the following multi-hop message dissemination schemes:

- A *simple flooding strategy*, where nodes flood the network by re-forwarding the messages they received. Nevertheless, we implement the following improvement for fair comparison: if a node receives an E-REQ message for an emergency already marked as solved (i.e. it has already received the corresponding E-REP message), it avoids to forward the E-REQ and immediately sends back the E-REP. Like the DTN-like state, every node waits for a random timing offset, uniformly distributed between $[0, 20]$ seconds before each retransmission.
- A *probabilistic scheme*, where the dissemination protocol works identically to the previous case, except that nodes decide to re-transmit a message with probability $q = 40\%$. For fair comparison, the probability is equal to p_{start} , i.e. the retransmit probability in the DTN phase used by LOCATE (Equation 5).
- *LOCATE, basic DTN*, which is our dissemination protocol, without the optimization to the DTN phase in terms of messages sent described in Section IV-B.
- *LOCATE with 802.11*, which is our dissemination protocol implemented over the popular 802.11 Wi-Fi technology, available on all the existing smartphone devices (assuming the ad-hoc mode is enabled). This is considered as reference for state-of-art phone-based ECS.

C. Metrics

We compute the following network metrics:

- *Emergency Resolution Time (ERT)*, defined as the average time which is needed to solve the emergency, and computed as the time interval from when the ES

broadcasts the E-REQ message the first time, to when the first corresponding E-REP message is received.

- *Emergency Resolution Ratio (ERR)*, defined as the percentage of cases in which the emergency is solved, i.e. the ES receives a E-REP message within a maximum time threshold E_{thr} . We set E_{thr} equal to 30 minutes.
- *Emergency Overhead (EO)*, defined as the total number of E-REQ messages exchanged among LoRa nodes during the entire simulation.

D. Results

In the first analysis, we vary the ratio of solvers, i.e. τ , on the x-axis, while keeping fixed the number of nodes, i.e. $n=40$. Figure 7(a) shows the ERT metric for the five schemes described above. It is easy to notice that: (i) the performance of all schemes improve when increasing the number of nodes able to send the E-REP message, as expected; (ii) the LoRa-based solutions guarantee much lower ERT values than Wi-Fi solutions thanks to the range expansion; (iii) the full LOCATE scheme greatly overcomes the *Flooding* and *Probabilistic* schemes, specially in critical scenarios where the number of solvers is low. Figure 7(b) shows the ERR metric over τ : the LOCATE scheme provides the highest probability of solving the emergency within the E_{thr} temporal threshold, for all the values of τ . For $\tau=30\%$, the ERR metric approaches the 100%. Figure 7(c) shows the EO metric, and shows that: (i) the DTN optimization greatly reduces the number of message sent within the network, and (ii) the improvement produced by LOCATE in terms of ERR and ERT does not come at the expense of the EO , since the number of messages sent in the network is comparable with the *Probabilistic* and *Flooding* schemes. In the second analysis, we vary the number of nodes, i.e. n , while keeping fixed the ratio of solvers, i.e. $\tau=15\%$. We do not show the LOCATE over Wi-Fi, due to the poor performance of the latter. Analogously to Figure 7(a), Figure 8(a) shows the ERT metric for the four protocols. Increasing n has a positive effect on the ERT metric, since there are more solvers in the neighborhood of the ES . However, we can see that the LOCATE scheme provides a considerable

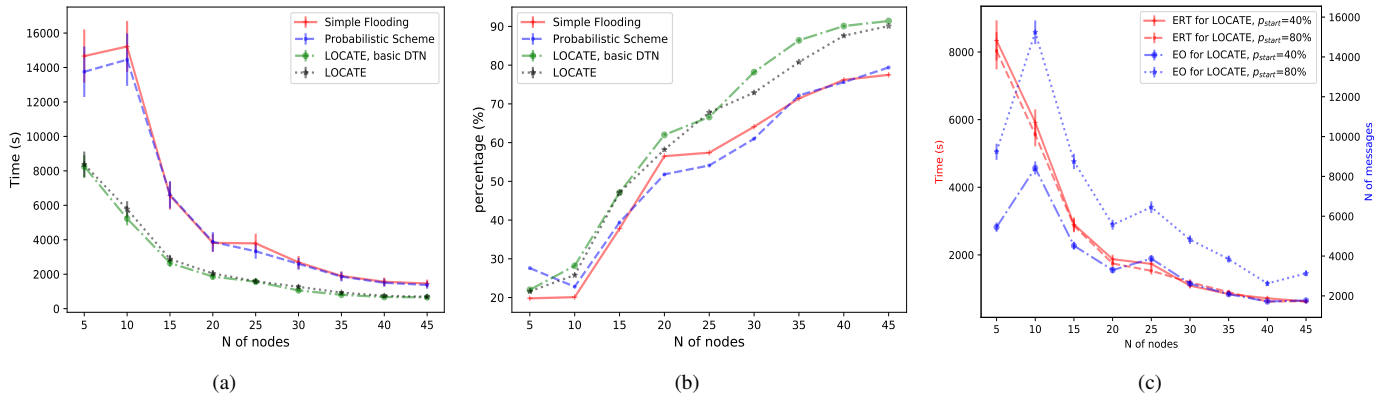


Fig. 8. The ERT, ERR and EO metrics for different values of n and $\tau=15\%$ are shown in Figures 8(a), 8(b) and 8(c) respectively.

delay reduction ($\sim 50\%$ for $n = 5$) even under low density scenarios, which are the most critical for the ECS deployment. Figure 8(b) depicts the ERR metric for the same analysis and confirms the same trend. Finally, Figure 8(c) investigates the impact of the p_{start} parameter of Equation 5 on the LOCATE performance. More specifically, we depict the ERT metric on the y-axis, and the EO metric on the y2-axis, for two different configurations of p_{start} , i.e. $p_{start}=40\%$ and $p_{start}=80\%$. Clearly, for $p_{start}=80\%$, the probability of retransmitting each message during the DTN phase is higher, and hence also the EO metric increases. However, the interesting result is that the two configurations of p_{start} guarantee almost identical performance in terms of ERT; in other words, the time for solving the emergency is only marginally affected by the probability to retransmit the E-REQ and E-REP messages. For this reason, we used $p_{start}=40\%$ in the analysis.

VI. CONCLUSION AND FUTURE WORKS

In this paper, we have investigated how to implement next-generation ECS able to leverage on the pervasiveness of smartphones, and on the excellent propagation characteristics of the LoRa technology in order to support data exchange even in highly critical scenarios. To this aim, we have proposed a novel mobile application, allowing the user to broadcast emergency requests with minimal, yet vital information on the LoRa link, and a novel dissemination protocol, allowing multi-hop spreading of the emergency requests over infrastructure-less phone networks. We have evaluated the LOCATE system through experimental measurements and extensive OMNeT++ simulations. Future works include: the analytical modeling of the dissemination protocol, the dynamic, distributed optimization of LoRa parameters on each LOCATE device, the experimental evaluation in rural scenarios.

REFERENCES

- [1] R. Ferrus, O. Sallent, G. Baldini and L. Goratti. Public safety communications. *IEEE VTM*: 7(2), pp. 54-61, 2012.
- [2] S. M. George, W. Zhou, H. Chenji, M. Won, Y. O. Lee, A. Pazarloglou, R. Stoleru and P. Barooah. DistressNet: a wireless ad hoc and sensor network architecture for situation management in disaster response. *IEEE Communication Magazine*: 48(3), pp. 128-136, 2010.
- [3] Gianluca Aloï, Orazio Briante, Marco Di Felice, Giuseppe Ruggeri, Stefano Savazzi. The SENSE-ME platform: Infrastructure-less smartphone connectivity and decentralized sensing for emergency management. *Pervasive and Mobile Computing*, 42(1), pp. 187-208, 2017.
- [4] The FireChat Application <http://opengarden.com/firechat/>
- [5] O. Mokryn, D. Karmi, A. Elkayam, T. Teller. Help Me: opportunistic smart rescue application and system. *Proc. of IEEE Med-Hoc-Net*, Ayia Napa, Cyprus, 2012.
- [6] M Di Felice, L. Bedogni, L. Bononi. The Emergency Direct Mobile App: Safety Message Dissemination over a Multi-Group Network of Smartphones using Wi-Fi Direct. *Proc. of ACM MobiWac*, La Valletta, Malta, 2016.
- [7] A. Asadi, Q. Wang, V. Mancuso. A survey on device-to-device communication in cellular networks *IEEE Communication Surveys & Tutorials*, 16(4), pp. 1801-1819, 2014.
- [8] U. Raza, P. Kulkarni and M. Sooriyabandara. Low Power Wide Area Networks: An Overview. *IEEE Communication Surveys and Tutorials*, 19(2), pp. 855-874, 2017.
- [9] LoRa Alliance: <https://www.lora-alliance.org/lorawan-white-papers>.
- [10] J. Petajajarvi, K. Mikhaylov, A. Roivaninen, T. Hanninen and M. Pettissalo. On the Coverage of LPWANs: Range Evaluation and Channel Attenuation Model for LoRa Technology. *Proc. of IEEE ITST*, Copenhagen, Denmark, 2015.
- [11] P. Jorke, et al. Urban Channel Models for Smart City IoT-Networks Based on Empirical Measurements of LoRa-links at 433 and 868 MHz. *Proc. of IEEE PIMRC*, Montreal, Canada, 2017.
- [12] O. Georgiu and U. Raza. LoW Power Wide Area Network Analysis: Can LoRa Scale?. *IEEE Wireless Communications Letters*, 6(2), pp. 162-164, 2017.
- [13] D. Magrin, M. Centenario and L. Vangelista. Performance Evaluation of LoRa Networks in a Smart City Scenario. *Proc. of IEEE ICC*, Paris, France, 2017.
- [14] C.-H. Liao, G. Zhu, D. Kuwabara, M. Suzuki and H. Morikawa. Multi-Hop LoRa Networks Enabled by Concurrent Transmission. *IEEE Access*, pp. 21430-21446, 2017.
- [15] J. So, D. Kim, H. Kim, H. Lee and S. Park. LoRACloud: LoRa Platform on OpenStack. *Proc. of IEEE Netsoft*, Seoul, South Korea, 2016.
- [16] W. Zhao, S. Lin, J. Han, R. Xu and L. Hou. Design and Implementation of Smart Irrigation System based on LoRa. *Proc. of IEEE Globecom Workshops*, Singapore, 2017.
- [17] K. Tzortzakos, K. Papafotis and P. P. Sotiriadis. Wireless Self Powered Environmental Monitoring System for Smart Cities based on LoRa. *Proc. of IEEE PACET*, Xanthi, Greece, 2017.
- [18] B. Sartori, S. Thielemans, K. Steenhaut, M. Bezunartea and A. Braeken. Enabling RPL multihop communications based on LoRa. *Proc. of IEEE WiMob*, Rome, Italy, 2017.
- [19] S.-T.Y. Wang, Y.-R. Chen and C.-H. Chang. Performance of LoRa-based IoT Applications on Campus. *Proc. of IEEE ICC*, Paris, France, 2017.
- [20] M. Slabicki, G. Premsankar and M. Di Francesco. Adaptive Configuration of LoRa Networks for Dense IoT Deployments. *Proc. of IEEE NOMS*, Taipei, Taiwan, 2018.